Step 4 Primary ROI Analysis

Overview

Metric	Value
Participant-sessions in Step 3 table	38
Primary-included participant-sessions	36
Primary valid HbO trials	1075
Manual Step 4 inclusion overrides	2
Global excluded participants	2
Logged Step 4 exclusions	3
Exploratory long channels tested	22

Global Participant Exclusions

The shared exclusion list from earlier steps remained active in Step 4. Files were retained for audit, but these participants were not used in the ROI analyses.

Metric	Value
PID006	Global downstream cluded from Step 1 ture downstream an sion 2025-11-12_002 second recording wi
PID037	Global downstream cluded from Step 1 future downstream exists but no fNIRS

Manual Step 4 Inclusions

The following sessions were included in Step 4 by explicit analyst override despite Step 3 review status. This deviation was requested and logged in the Step 4 specification file.

Metric	V
PID025 / 2025-11-25_002	U 20 V
PID034 / 2025-11-27_003	U 20 V

Primary Dissociation Test

n	Front A	Front C	Back A	Back C	Mean Δ	t	df	p	95\% CI	c
36	-0.0000	-0.0000	-0.0000	-0.0000	0.0000	1.3547	35	0.184185	[-0.0000, 0.0000]	0.2258

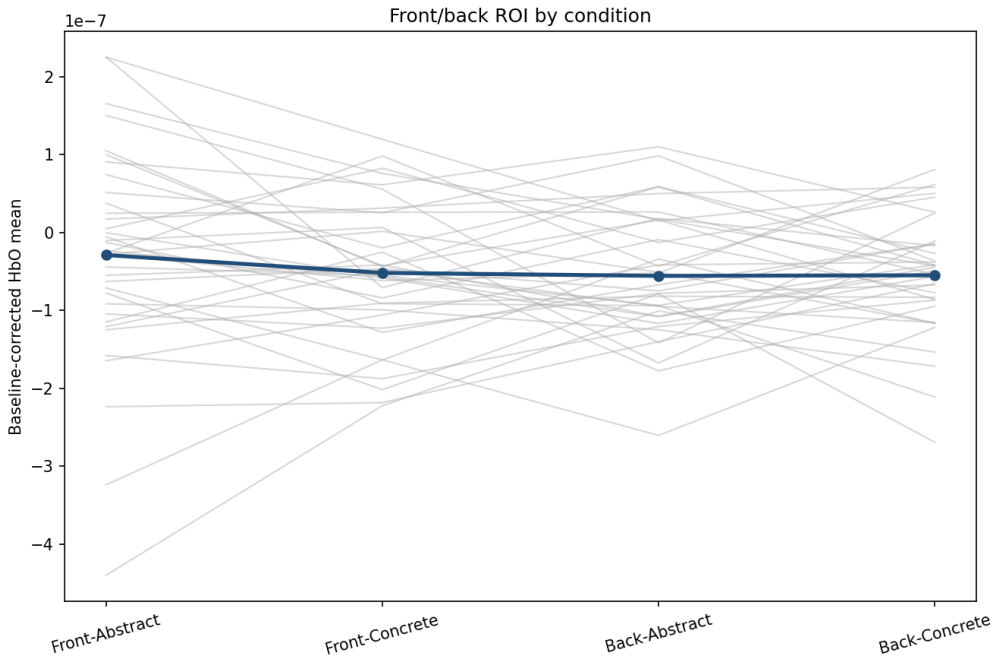

Figure 1: Participant-level four-cell HbO summary for the front and back ROIs.

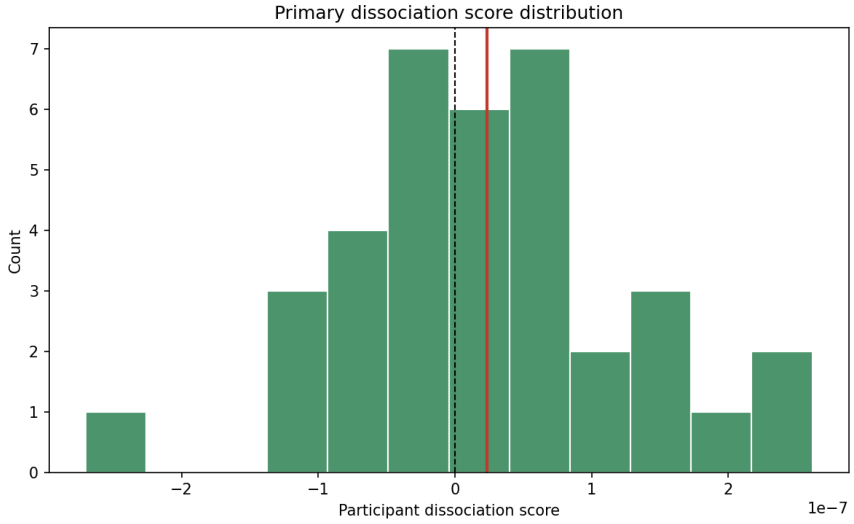

Figure 2: Distribution of the participant-level primary dissociation scores.

Planned Follow-Up Tests

ROI	n	Mean diff	t	df	p	Holm p	95\% CI	d$_z$
Front	36	0.0000	1.4117	35	0.166864	0.333728	[-0.0000, 0.0000]	0.2353
Back	36	-0.0000	-0.0496	35	0.960692	0.960692	[-0.0000, 0.0000]	-0.0083

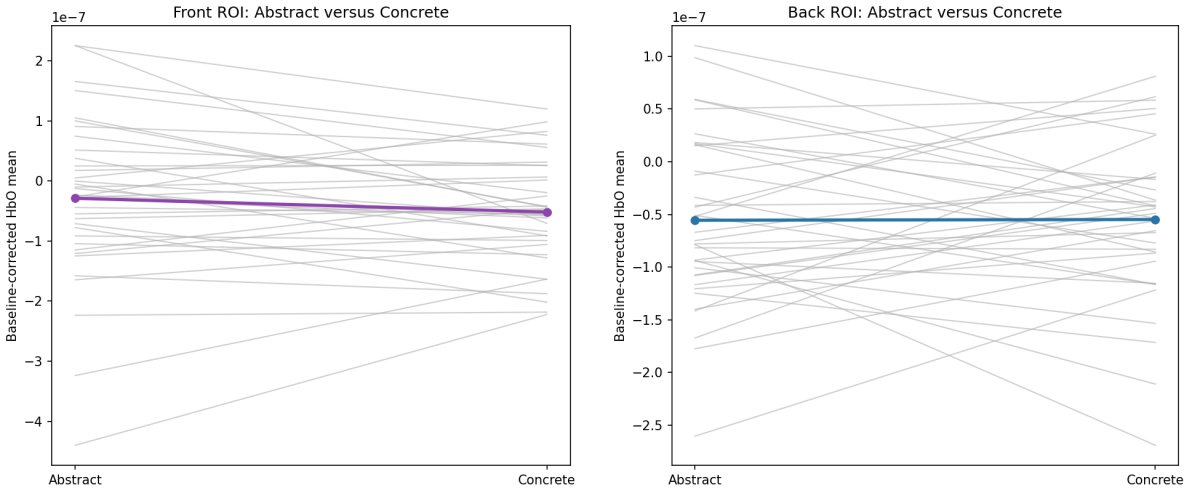

Figure 3: Planned paired comparisons for the front and back ROIs.

Secondary Analyses

Metric	Value
Secondary HbR primary-style p-value	0.838320
Smallest corrected exploratory channel p-value	0.355121

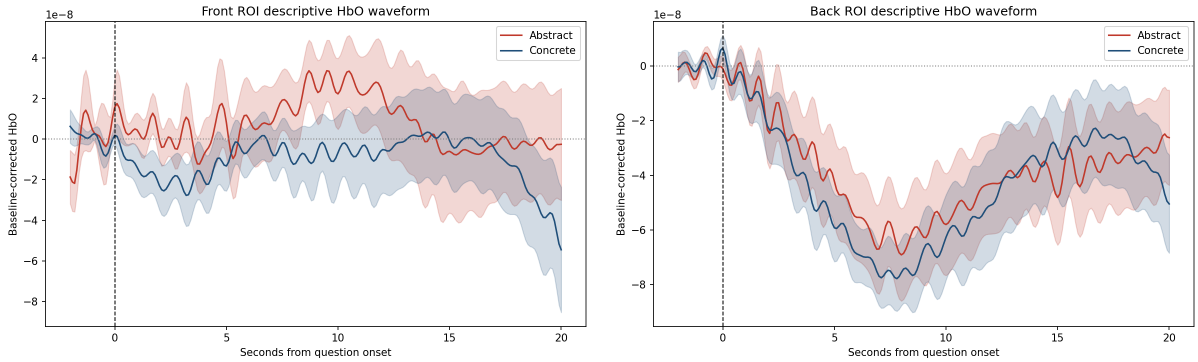

Figure 4: Descriptive HbO waveforms aligned to question onset for the front and back ROIs.

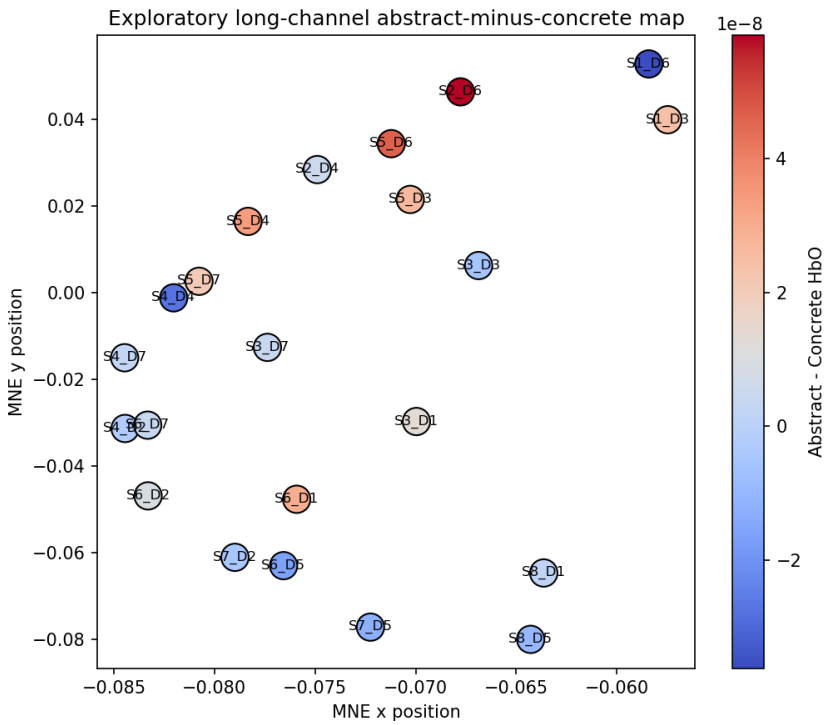

Figure 5: Exploratory long-channel abstract-minus-concrete HbO effects.

Final Conclusion

The pre-specified primary analysis did not show a statistically significant front-versus-back dissociation between abstract and concrete questions (t(35)=1.3547, p=0.184185, mean dissociation=0.0000, 95